

Preserving physical information in transferable wind-power event measurements

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Abstract

Wind-power events connect atmospheric variability to operational decisions, but their useful information depends on measurement and representation. We introduce a protocol-aware framework linking structural survival, independent physical tracking and forecast-to-cost validation across eight archives, 340 turbines and seventeen detector configurations. Ordered trajectories preserve transferable structure among matched, encodable events, while polarity protection restores a distinction erased by angular encoding. Reconstructing independently measured wind trajectories reveals where temporal information matters most. For power-defined compound events, scale-restored trajectories reduce reconstruction error by 11.7% and 24.5% against strong scalar controls at two light detection and ranging (LiDAR) instruments, with positive paired seven-day confidence intervals. Internal-geometry error falls by 9.6–26.9% across both instruments and a French WindCube population. Separately, historical shape and issued weather inputs support an 18.2% reduction in simulated four-hour accuracy charges relative to persistence under a documented Jiangsu rule. These findings establish event complexity and decision objectives as guides to representation design, connecting transferable structure to physical interpretation and economic evaluation.

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1. Introduction

Wind-power forecasting turns atmospheric variability into information for reserve scheduling, trading and flexible operation. Physical, statistical and learning-based methods have improved this conversion, while probabilistic forecasting has made forecast uncertainty part of the operational question [1, 2, 3]. Dynamic events provide an intermediate description between a continuous trajectory and an action: they locate when production changes, characterize how it changes and identify the time over which a response may be useful.

The event itself is a measurement choice. Ramp research has defined changes through amplitude, duration, timing and local trend [4, 5]. Swinging Door and optimized segmentation methods construct interval boundaries, while threshold, historical-tail and ordinal approaches emphasize different aspects of variability [6, 7, 8, 9, 10]. A large excursion that returns to its initial level illustrates the distinction: it has a small net change and a substantial internal dynamic structure. The scientific question is which of these structures remains recognizable when the measurement protocol changes.

Modern forecasting makes this question more consequential. Event grouping, stochastic scenarios, attention, transfer learning and multiscale models offer increasingly expressive representations of production changes [11, 12, 13, 14, 15, 16]. Recent event-aware and probabilistic wind-power models connect architectural improvements to operational prediction [17, 18]. Time-series representations also range from ordered coordinates to image encodings and learned temporal features [19, 20, 21, 22]. Their comparison therefore requires a shared definition of the process being represented, as well as a test of the information that the representation preserves.

Independent atmospheric measurements supply that test. Weather regimes, coupled wind-farm dynamics and spatial inflow variations all influence production trajectories [23, 24, 25]. Large-farm evaluations and numerical-weather-prediction studies demonstrate the importance of weather-model configuration, resolution and timing [26, 27, 28]. Event structure can consequently be evaluated in two complementary directions: weather information can predict production changes, and observed production structure can support probabilistic identification of the associated wind process. These directions connect event measurement to physically interpretable forecasting.

Here we organize wind-power event evaluation into four layers: detection, matching, representation and evaluation–decision (Fig. 1). Detection defines intervals; matching establishes correspondence; representation and clustering describe

their structure; evaluation tests their physical and operational use. Three criteria connect these layers. Structural survival measures partition agreement among corresponding events; coverage is the fraction of eligible events represented in a match, reported separately for each side. External-process tracking tests whether the representation identifies independently measured wind dynamics. Forecast-to-cost validation evaluates available information under the timing and charging rule of a decision. Chinese and European archives share the same physical-time interface, and a reserved later period tests frozen representations across time.

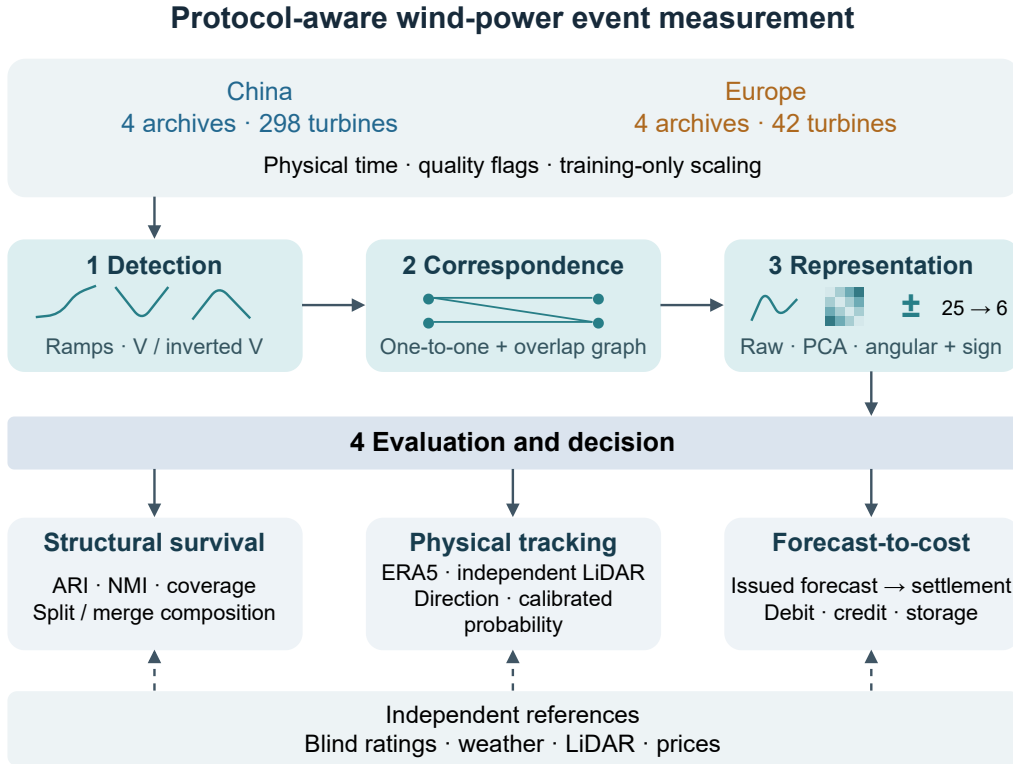


Figure 1: Four layers connect eight wind-farm archives (340 turbines) to three validation criteria. Detection defines ramp and compound intervals; correspondence uses one-to-one matches and split/merge overlap graphs; representations retain trajectory or angular information. Structural survival, external-process tracking and forecast-to-cost tests evaluate progressively different uses. Miniature trajectories, overlap links and the angular field are schematic. China and Europe identify data regions; panel fills distinguish measurement from evaluation. Dashed arrows supply independent reference evidence. Hill 2021 and 2026 are additional periods from the same farm.

We establish a task-aware approach to event measurement. Matched-event

comparisons identify structures that persist across detection protocols, and a localization benchmark tests how scores become event intervals. A global-sign ambiguity in angular encoding provides an explicit example of information loss and recovery. We then reconstruct complete, independently measured wind trajectories to test information beyond endpoint direction. This reveals a complexity-dependent benefit: compound changes make fuller use of ordered temporal information than primitive events. Finally, a native-resolution forecasting track connects historical shapes and issued weather forecasts to accuracy charges under a documented regional rule. The framework therefore supplies both a measurement interface and an empirical design principle: preserve the information required by the event’s physical complexity and the intended decision.

2. Results

2.1. A common physical-time interface supports transferable event structure

The harmonized catalogue contains 4,542,258 candidate intervals and 3,226,925 complete 25-coordinate shapes from seven archives and 333 turbines. Basic detector intervals and composite V-shaped or inverted-V episodes form separate event levels. Seventeen detector configurations share a 30-min grid, physical lags of 60, 120 and 240 min, a 24-h historical window and two hours of context on each side. Fig. 2 applies these rules to the same measured production trajectory, making protocol-dependent event boundaries visible before the aggregate comparison.

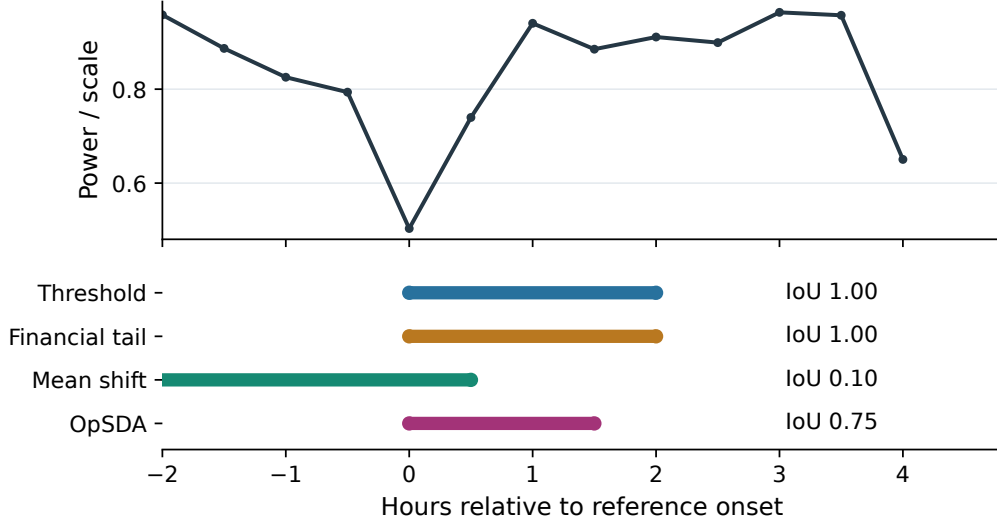


Figure 2: The same observed trajectory yields different event boundaries. The lower tracks show the four detector intervals on the same physical-time axis; threshold and financial-tail boundaries coincide in this example. IoU is measured against the threshold reference. Track labels replace an overlapping legend.

Pizhou and Greece each supply 9,925 training events under a matched sampling budget. Six representations, three clustering algorithms, three cluster counts and three seeds yield 324 fitted source models and 648 source–target evaluations. The primary analysis fixes K-means at $k=4$; $k=2$ and $k=6$ test coarser and finer partitions. On the configuration pairs shared by all six primary representations and all three seeds, raw25 median adjusted Rand index (ARI) ranges from 0.714 to 0.817 across the seven archives. ARI measures agreement between cluster partitions after correcting for random agreement: 1 denotes identical partitions up to relabelling, 0 the chance expectation, and negative values less agreement than expected by chance. A supported configuration pair contains at least 100 matched event pairs; a primary informative pair additionally has nonconstant partitions in both protocols. Greece provides 66 such supported configuration pairs and median ARI 0.817; the reserved Hill 2021 period provides 112 and ARI 0.711. An event pair is one matched interval from each protocol, whereas a configuration pair is the pair of detector settings being compared.

Archive	Turbines	Candidates	Complete shapes	Validation + test matches
Pizhou	33	880,613	606,140	611,159
Suining	14	287,061	227,472	243,765
Yandun	117	669,958	539,542	1,078,205
La Haute Borne	4	83,622	75,402	74,258
Hill of Towie 2020	21	518,078	468,272	478,528
Greece	10	64,286	54,249	40,164
SDWPF	134	2,038,640	1,255,848	2,516,079
Hill of Towie 2021	21	182,660	168,236	449,388

Match counts in this catalogue table include both event hierarchies and the validation and test periods. The structural comparison below uses basic-interval test events. Supported pairs are detector-configuration pairs with at least 100 matched events and informative partitions common to the representations. Coverage is the mean left/right catalogue fraction across all configuration comparisons, reported separately from agreement on the supported subset.

Archive	Supported pairs	Raw25 ARI	Left coverage	Right coverage
Pizhou	105	0.747	34.5%	30.9%
Suining	98	0.724	34.5%	32.5%
Yandun	119	0.714	34.6%	35.5%
La Haute Borne	89	0.785	32.2%	32.3%
Hill of Towie 2020	104	0.717	34.8%	32.8%
Greece	66	0.817	31.2%	29.7%
SDWPF	119	0.745	33.5%	31.1%
Hill of Towie 2021	112	0.711	33.4%	33.2%

The 2021 evaluation adds a period from the same 21 Hill of Towie turbines. Its 168,236 complete shapes and 367,102 basic-interval test matches extend the

temporal test while retaining the original training transformations. Coverage is recorded separately for each side of every match, so the comparison has an explicit population as well as an agreement score.

A metric sensitivity reuses the frozen Pizhou K-means models at $k=4$ and seed 41. Raw25 ranks above ordinary angular principal component analysis (PCA) and protected-polarity angular PCA at all seven archives and the reserved period under adjusted mutual information (AMI), the Fowlkes–Mallows index and variation of information (VI), as well as ARI and normalized mutual information (NMI). The single-seed comparison uses the same supported event pairs for all three representations. Agreement across these metrics strengthens the interpretation of raw25 as a stable structural reference, while the following experiments evaluate the distinct physical information retained by angular representations.

Raw-PCA6 is a compression control derived from the same 25-point trajectory (Fig. 3). It retains 93.4% of standardized Pizhou training variance. Across the three seeds, the ARI between its Pizhou test partition and the corresponding raw25 partition is 0.985–0.998. Cross-protocol site medians differ by -0.0022 to 0.0027 across the seven primary archives (Fig. 4), with full-precision configuration differences archived alongside the summaries. The retained low-dimensional trajectory structure explains the close values.

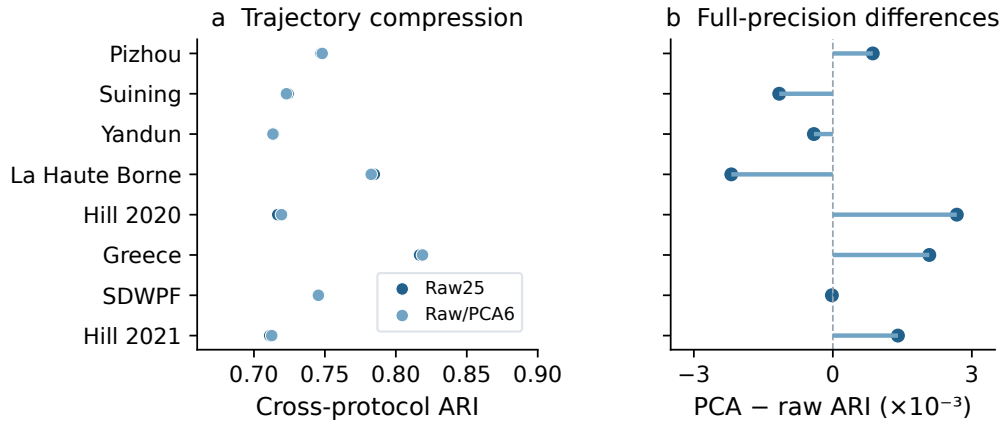


Figure 3: Raw/PCA6 provides a compression control for the same 25-point trajectory. Panel a shows current common-support v18 results; panel b resolves differences hidden by two-decimal rounding. PCA retains 93.4% of standardized training variance. The separate event-label partition comparison is reported in the accompanying data; it differs from cross-protocol ARI.

The primary $k=4$ comparison resolves four empirical groups rather than four

prescribed physical classes. Cluster-count sensitivity is reported explicitly in Supplementary Table S57. In the common-support Pizhou comparison, raw25 median ARI is 0.930, 0.747 and 0.653 at $k=2, 4$ and 6, respectively; the corresponding ordinary GAF/PCA values are 0.147, 0.227 and 0.284. Raw25 remains above angular PCA at all three cluster counts in Pizhou, Greece and Hill 2021. At $k=6$, statistics9 reaches ARI 0.835 in Greece and 0.651 in Hill 2021, exceeding raw25 at 0.735 and 0.587. These results identify a resolution-dependent comparison: coarse trajectory groupings transfer strongly, while finer scalar partitions can preserve additional distinctions. The complete six-representation $k=2/6$ contrasts concern these three evaluation populations; raw25-only $k=2/6$ transfer is also reported for the other archives. The corresponding primary comparison is shown in Fig. 4.

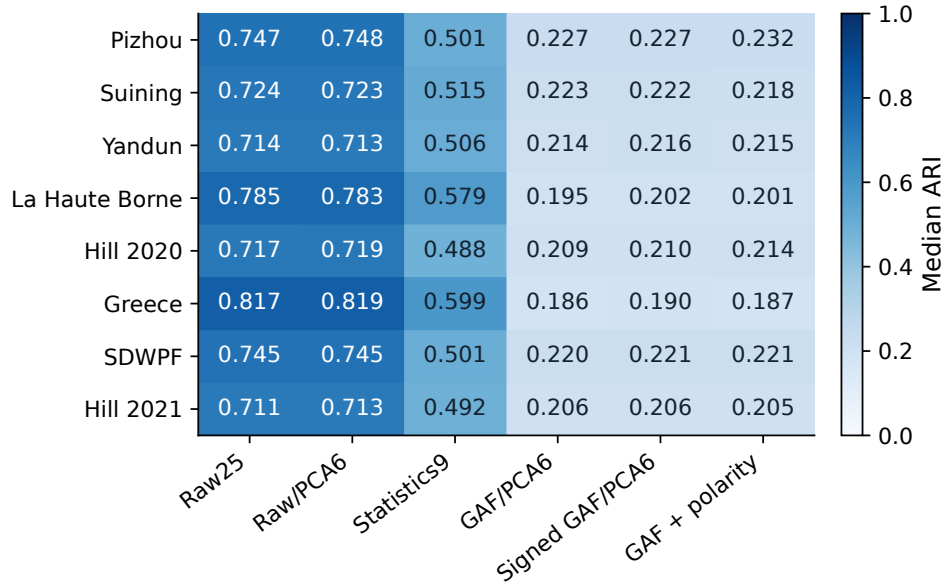


Figure 4: Structural survival across seven archives and the Hill 2021 temporal holdout. Darker blue represents higher ARI; label contrast follows cell luminance. Each cell is the median over common detector-configuration pairs after the median over three training seeds. Raw/PCA6 and the angular variants are related ablations, whose differences are resolved in Fig. 3 and the full-precision data.

2.2. Protocol perturbations identify a stable core and complementary geometry

An overlap graph extends correspondence to event splitting and merging. At $\text{IoU} \geq 0.5$, 6.6–9.5% of primitive-event overlap components contain more than one event on at least one side. Including primitive and composite V/inverted-V intervals raises this share to 19.8–36.7% across the seven archives and Hill 2021. This

quantifies the additional correspondence revealed when a single compound change is represented as several detector intervals. One-to-one pairing supports event-level partition agreement, whereas the overlap graph preserves the split/merge composition of shared episodes (Supplementary Table S58 and Fig. S1).

Changing an amplitude threshold changes the event population and its boundaries. Across all seven archives, thresholds of 0.10, 0.15, 0.25 and 0.30 were compared with the 0.20 reference using frozen amplitude scales and prototypes. Financial-tail probabilities of 0.01, 0.025 and 0.10 were similarly compared with the 0.05 reference. Crossing these seven perturbations with three lags (60, 120 and 240 min) and seven sites gives 147 site–parameter–configuration comparisons, each carrying one-to-one event matches and separate left/right coverage. These tests identify which measured structures survive a controlled change to the detector; the sampling examples in Fig. 5 show the same physical-time principle under changes to the observation grid.

Sampling provides a second physical perturbation. The same-context Greek 30-versus-60-min comparison gives a median ARI of 0.942, with median minimum-side coverage of 0.484. Greek 10-versus-30-min values are 0.866 and 0.551. At Yandun, 15-versus-30-min values are 0.761 and 0.708. The added 30-versus-60-min comparisons give median ARI 0.857 for La Haute Borne and 0.794 for Hill of Towie, with median matched-side coverage 0.396 and 0.412 among configurations with at least one match. Fig. 5 displays the full detector-configuration distributions behind these summaries. Agreement and coverage have distinct meanings: a retained structure can remain recognizable among matched events while the observation grid changes which intervals can be compared.

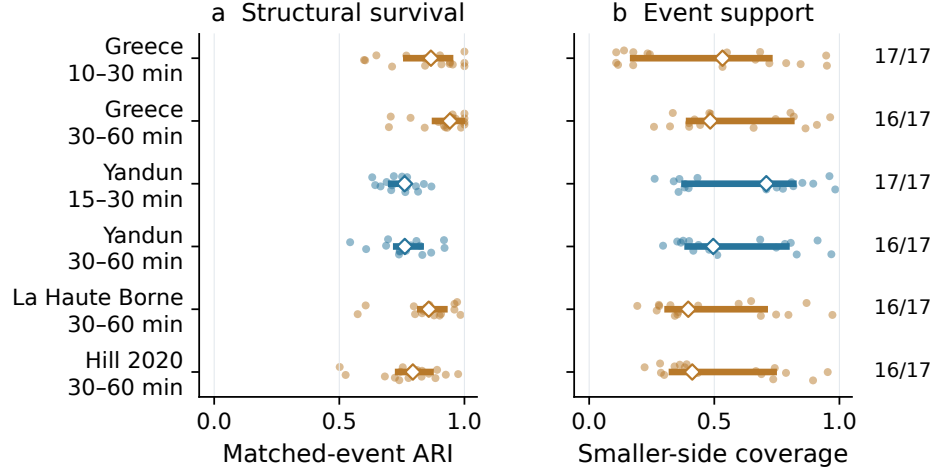


Figure 5: Sampling sensitivity now includes Greece, Yandun, La Haute Borne and Hill of Towie. Dots are detector configurations; diamonds and bars are medians and interquartile ranges across informative configurations. Counts give configurations with at least one match out of 17. Scale, prototype, split and physical context remain fixed. All eight grid comparisons, including Greece 10/60 and Yandun 15/60, remain in the data table; plotted comparisons connect native/30-minute and common 30/60-minute grids.

The content of that structure is revealed by conditioning. Fig. 6 reports the conditioned information after accounting for protocol pair, direction, amplitude, duration and pre-event power strata. Angular partitions retain about 0.40–0.46 nats of shared information above within-stratum permutation across the archives. Here a nat is the natural-log unit of information ($1 \text{ nat} = 1/\ln(2) \text{ bits}$). The corresponding raw25 values are about 0.02–0.04 nats. Ordered-coordinate partitions allocate much of their information to polarity, whereas angular partitions preserve a larger proportion of shared geometry within the conditioned populations. We use stable core to mean the raw25 partition structure retained across detector protocols, and complementary geometry to mean the shared angular-partition information remaining after conditioning on these physical descriptors.

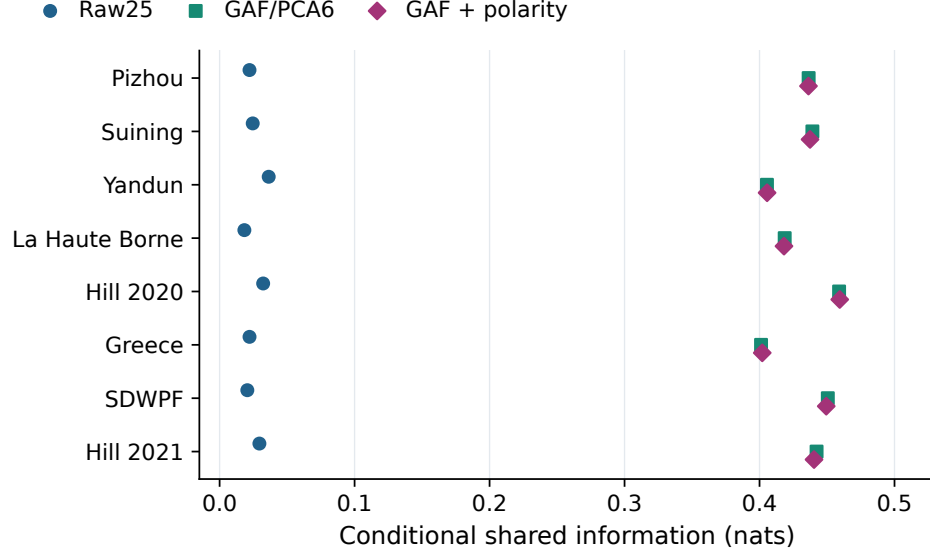


Figure 6: Shared geometry after conditioning on protocol pair, direction, amplitude, duration and pre-window power. Values subtract the mean within-stratum permutation reference. The near-overlap of the angular variants shows similar geometry under these controls. Protected polarity targets directional information, tested with independent wind measurements in the following figures; it is not selected to maximize this conditioned score.

2.3. Protected polarity preserves a physically meaningful distinction

For the signed-domain Gramian angular summation field (GAF) used here, a path and its sign reversal produce the same image. Fig. 7 shows the paired paths and Fig. 8 shows their identical angular field. This identity gives a precise intervention: one sign bit at a nonzero reference coordinate resolves the ambiguity of the full field. For a compact representation, we retain five coordinates from principal component analysis (PCA) and reserve a sixth coordinate for polarity. This protects the directional distinction through compression; exact recovery of the entire path is a property of the full field plus sign, whereas the compressed representation is evaluated empirically. The next experiment tests the physical value of retaining that distinction.

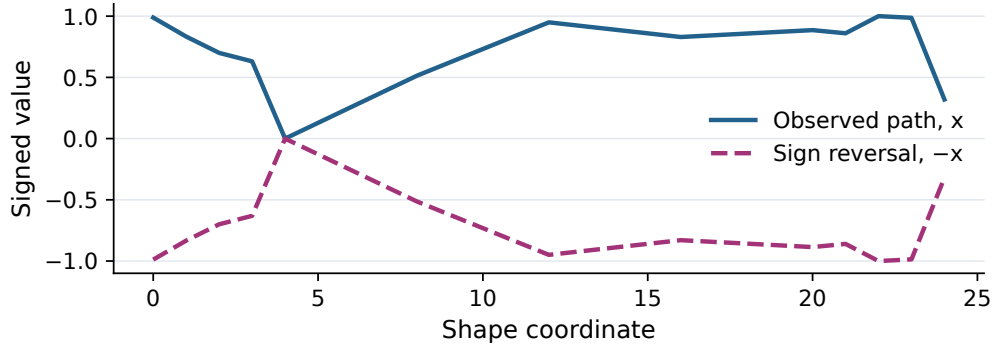


Figure 7: An observed 25-coordinate path and its controlled global sign reversal. These two paths have opposite polarity and the same signed-domain Gramian angular summation field.

The empirical comparison separates preserving a coordinate from appending an input channel. Appending signed coordinates before PCA exposes them to the same compression as the image. Reserving a polarity coordinate after PCA keeps the chosen distinction explicit. Both representations remain compact, but they encode different guarantees about directional information.

The protected sign refers to the largest departure across the full contextual trajectory, whereas the scalar direction refers to the net change between event endpoints. These signs coincide in 90.8–97.2% of events in the five populations examined for anchor sensitivity. The remaining 2.8–9.2% distinguish the dominant contextual excursion from the net event direction (Supplementary Table S74). This relation motivates two separate tests: recovering a sign erased by angular encoding, and measuring its additional predictive value when scalar direction is already supplied.

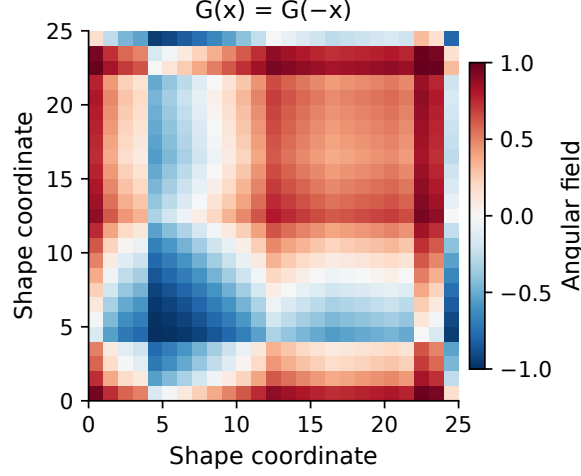


Figure 8: The common Gramian angular summation field for the two paths in Fig. 7. The identity $G(x)=G(-x)$ motivates retaining a sign coordinate alongside the compressed angular representation.

In the primary fit, the protected polarity coordinate has a variance share bounded above by approximately 0.225% in the six-dimensional encoding. Its small geometric weight explains the similar cluster partitions; its scientific role is to retain a direction that can be read by a supervised physical head.

The compact comparison contains six-dimensional ordinary GAF/PCA, signed-channel GAF/PCA, protected-polarity GAF/PCA and raw/PCA. Raw25 and the nine-statistic representation provide explicit higher-dimensional references. All variants use the same training events and weather targets, with shared downstream model-selection budgets. This comparison tests whether an explicit polarity coordinate supplies useful directional information at a fixed six-dimensional representation budget.

Five regional-weather evaluations relate these event representations to signed 100-m wind-speed change. At Hill of Towie, the protected-polarity representation reduces held-out signed log loss from 0.379 to 0.345, a 9.03% reduction, with a seven-day block interval of 6.35–10.96%. Decomposing the same probability score into occurrence and direction components explains the gain. Conditional wind-tendency AUROC rises from 0.538 to 0.932, while the occurrence component changes little. The intervention principally restores the distinction that its mathematical construction was designed to preserve.

The five evaluations are Pizhou, Suining, Yandun, La Haute Borne and Hill of

Towie. At Hill, the occurrence component is 0.307 for GAF/PCA and 0.311 for protected polarity, whereas the direction component falls from 0.073 to 0.034. The gain is therefore directional information rather than a change in event frequency.

Continuous shape and scalar metadata provide a second route to process information. Adding raw25 to direction, amplitude, duration, initial power and pre-window power reduces signed regional-weather log loss pointwise at all five sites. The complete block intervals and the corresponding hard-cluster-label probes are reported together in the Supplementary Information. This comparison separates the information in a trajectory from the coarser information carried by a four-class label.

Prototype geometry provides a compact bridge between these descriptions. Adding distances to the four frozen raw25 prototypes improves signed-weather log loss pointwise at all five sites. At Yandun, the four-distance input reduces log loss by 0.0152 nats beyond the scalar controls. Retaining graded location within the learned structure therefore supports a physical-probability interface while preserving a compact four-coordinate description.

2.4. Independent LiDAR validates directional information recovery

To test the external validity of protected polarity, we use independent 2026 LiDAR measurements. Fig. 9 places the two instruments on one ROC canvas. A ground-based instrument near T11 measures horizontal wind near hub height, and a nacelle instrument on T07 measures upstream flow at a 208-m range. Applying the frozen Hill-2020 physical probes gives 5,106 and 2,627 matched events with complete wind support at the primary clock alignment. Protected polarity reduces signed log loss relative to angular PCA by 7.21% at T11 and 9.20% at T07. The respective seven-day block intervals are 5.37–8.85% and 7.42–10.72%; both gains also retain their direction under the predeclared ± 10 -min clock checks.

A separate chronological field-calibration experiment translates the source scores into local probabilities. Calibration training ends before 14 March, validation ends before 7 April, and testing uses later events with their full contexts contained in the test period. The three-class test contains 1,064 T11 and 444 T07 events. Directional AUROC uses the 705 and 285 events with measured wind increase or decrease, respectively, occupying four seven-day blocks at each instrument. Protected polarity attains AUROC 0.890 (95% block interval 0.822–0.974) and 0.905 (0.808–0.947), compared with 0.494 (0.468–0.544) and 0.477 (0.321–0.599) for angular PCA. Three-class signed log loss decreases by 16.42% and 16.11%, respectively. The independent measured outcome thus identifies a useful physical distinction in the event representation. ARI and

AUROC answer different questions: ARI measures agreement between partitions of matched events; AUROC measures discrimination of independently measured wind tendencies.

Scalar descriptors provide an alternative route to directional information. On the same calibrated LiDAR events, the protected representation exceeds scalar-only direction AUROC by 0.0320 at T11 and 0.0178 at T07; paired seven-day intervals are -0.0207 to 0.0587 and -0.1561 to 0.0669, respectively. These comparisons use four occupied seven-day blocks per instrument, preserving all event pairs in each resample. A factorial comparison then holds five scalar descriptors and the first five angular principal components fixed, and adds the polarity coordinate separately. The two LiDAR tests show comparable conditional discrimination with and without this added coordinate (Supplementary Tables S65–S68). The supported mechanism is information protection: polarity restores a distinction erased by angular encoding, while its incremental value depends on whether accompanying descriptors already supply that distinction.

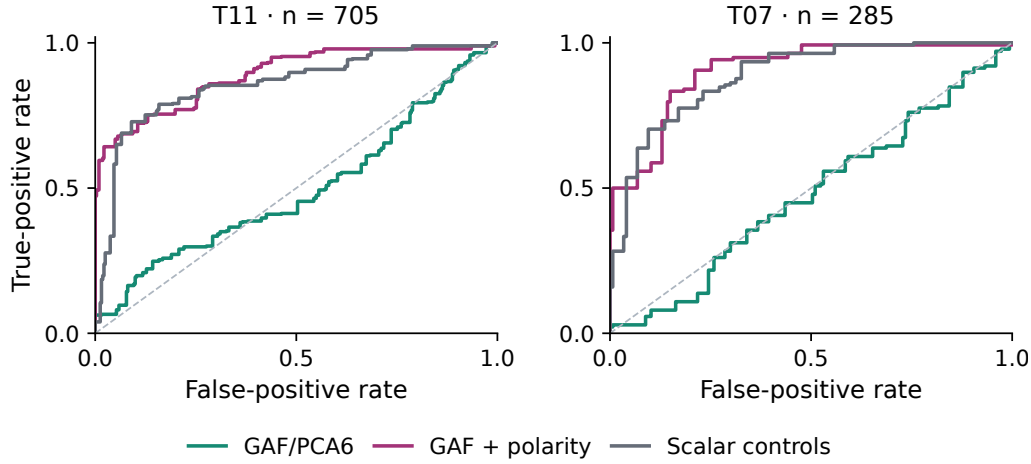


Figure 9: Independent LiDAR direction discrimination after chronological calibration. T11 and T07 occupy the left and right panels, with 705 and 285 measured increase/decrease events. Small-net-change observations are retained in the separate three-class score. Both instruments use the same representation colors and axes.

This experiment answers the reverse-identification question directly: an observed power-event description can support a probability for an externally measured wind-speed tendency. Section 2.6 extends this endpoint question to the complete wind trajectory, including changes that reverse within an event. Its forecast-

ing counterpart uses information available at issue time, as tested in the forecast-to-cost experiment in Section 2.8.

2.5. *A second instrumented field archive preserves the event interface*

SMARTEOLE supplies a second public instrumented wind farm: seven Sen-vion MM82 turbines, 1-min supervisory control and data acquisition (SCADA), a ground WindCube profiler and a yaw-control log. Fig. 10 combines its physical transfer with the resampled LiDAR distributions. Applying the frozen Pizhou K-means ($k=4$, seed 41) transformations after complete 30-min aggregation produces 77 supported turbine-by-configuration-pair rows, containing 8,985 matched primitive-event pairs per representation summary. Raw25 reaches median ARI 1.000 on this high-support subset; 45 of the 77 rows have ARI=1.000, while the pair-weighted ARI is 0.964 and the all-row pair-weighted ARI is 0.887 across 852 rows. GAF/PCA and protected polarity reach 0.605 and 0.597 on the same supported-row summary. Here agreement refers to the retained partition on matched events. Reporting the support rule and weighted diagnostics makes the transfer result transparent: the high-support raw25 median is a concentrated structural-transfer signal, not an identity claim for every detector pair. The structural result establishes a working transfer of the event ledger and one-to-one correspondence interface to a different field archive.

The same frozen Pizhou physical heads use WindCube 80-m endpoint wind-speed changes as an independent outcome. Across the seven turbines, models combining scalar event descriptors with raw25 attain direction AUROC from 0.902 to 0.926. Corresponding angular and protected-polarity models retain the same scalar inputs, including direction, to evaluate shape information under a common control set. The SMARTEOLE provider removes rows with at least one stopped turbine, so this experiment identifies structure and wind-process information during the supplied operating population.

The scalar-controlled factorial test identifies an external setting with a positive conditional increment. Pooling the seven turbines while resampling their shared calendar blocks, scalar controls, scalar plus five angular components, scalar plus polarity, and the combined representation attain direction AUROC 0.8579, 0.8748, 0.8862 and 0.8835, respectively. Adding polarity to the same scalar-plus-angular model gives an AUROC gain of 0.0087 (paired seven-day interval 0.0039–0.0154) and a three-class log-loss reduction of 0.0135 nats (0.0091–0.0186). The comparison uses 13,931 eligible events, including 9,265 increase/decrease events, in 15 shared seven-day blocks. Relative to the independently randomized-coordinate control, the AUROC difference is 0.0026 (–0.0026 to 0.0089). Test-only reversal

of the actual sign reduces AUROC by 0.0490 (0.0340–0.0650), whereas consistent reversal during both fitting and testing preserves the encoding. Together these controls separate directional dependence from a generic benefit of adding one feature.



Figure 10: External physical validation across two instrumented fields. Top: standalone angular/protected encodings and a scalar-only control; empirical directional-AUROC distributions from 2,000 resamples of four occupied seven-day blocks per Hill instrument; bars show percentile intervals and dots medians. Bottom: seven turbine responses paired with the same SMARTEOLE WindCube reference, with scalar controls shared across representations. SMARTEOLE structural agreement has a raw25 median of 1.000 across 77 high-support rows, of which 45 equal 1.000; pair-weighted ARI is 0.964 and all-row pair-weighted ARI is 0.887. Structural row distributions are supplied separately.

2.6. Ordered trajectories recover the physical structure of compound events

The physical value of event geometry becomes clearest when a process changes direction within its measured interval. We therefore reconstruct seventeen points along an independently measured wind-speed trajectory, rather than reducing that trajectory to its endpoint tendency. The experiment combines primitive and compound power intervals lasting 60–240 min, with continuous wind support and complete contexts inside each chronological split. After identical-interval deduplication, T11, T07 and SMARTEOLE contribute 2,304, 1,038 and 7,040 intervals, respectively; their test populations contain 507, 170 and 449 intervals. Models and observation support are selected on validation data and frozen before evaluation.

Scale-restored ordered trajectories provide their largest full-process gains on compound events. Relative to validation-selected scalar and scalar-residual controls using the same native 10-min inputs, wind-trajectory root mean square error (RMSE) falls by 11.67% at T11 and 24.45% at T07 (paired seven-day 95% intervals 4.91–21.43% and 16.78–41.87%). These groups contain 165 and 38 intervals identified as compound by the power catalogue, independently of the wind outcome. The corresponding primitive-only changes are –0.72% and –1.30%, with intervals spanning zero. At SMARTEOLE, the compound full-trajectory reduction is 5.81%, with a wider interval of –9.94–7.27%. The result locates the strongest incremental value in the internal evolution of compound changes, where endpoints and scalar summaries provide a less complete description.

Physical reference	n	Compound RMSE reduction (%)	All-event geometry reduction (%)
T11 LiDAR	165	11.67 [4.91, 21.43]	11.09 [6.21, 19.86]
T07 LiDAR	38	24.45 [16.78, 41.87]	26.94 [17.80, 40.66]
SMARTEOLE	42	5.81 [–9.94, 7.27]	9.58 [9.03, 10.76]
WindCube			

Here n counts compound test intervals; brackets give paired seven-day 95% intervals. Internal geometry is the wind trajectory remaining after removal of its endpoint chord; its all-event denominators are 507, 170 and 449, respectively. Each population occupies four seven-day test blocks. Complete all-event, primitive-only and 3/14-day results accompany these comparisons in Supplementary Tables S75–S78.

Duration alone does not explain the Hill contrast. Standardizing compound and primitive-only groups to a common duration distribution leaves compound-minus-primitive gain differences of 10.65 percentage points at T11 and 28.16 at

T07, with intervals of 1.49–17.98 and 19.74–33.88 percentage points. The weights depend on power-event duration and shared support. Thus the advantage persists beyond the longer duration often associated with a two-leg episode.

Internal wind geometry improves across all three reference populations. Removing each trajectory’s endpoint chord exposes intermediate accelerations and reversals: reconstruction RMSE falls by 11.09%, 26.94% and 9.58% at T11, T07 and SMARTEOLE, respectively, with positive seven-day intervals. Full-trajectory reductions over all eligible events are 4.50%, 4.47% and 1.72%. This contrast separates the information recovered inside an event from the coarser endpoint trend. Perturbing only the temporal coordinates of the frozen decoder, while preserving scalar descriptors and endpoints, raises geometry RMSE in every population (Supplementary Table S79). Ordered correspondence therefore carries useful measured-process information beyond the supplied summaries.

These results yield a practical representation principle: retain native temporal support and ordered, scale-aware trajectories when the scientific target includes compound dynamics. The validation-selected decoder is regularized linear regression in all three populations. The gain is therefore achieved through the observation and representation interface, with the complete nonlinear and compressed-encoding comparisons retained in Supplementary S26.

2.7. Controlled localization connects event scores to detection

Structural survival and physical tracking answer different questions from event localization. We therefore retain the earlier controlled episode benchmark as a task-level validation. Synthetic 96-point trajectories contain known event support, direction, amplitude and shape. TimesNet, KAN-AD (Kolmogorov–Arnold network anomaly detection), a temporal convolutional network (TCN) autoencoder and a positional Transformer autoencoder receive the same training sequences and score-to-interval budget. Threshold calibration and event grouping are evaluated as separate output steps.

At fixed model scores, calibration followed by grouping increases F1 from 0.263 to 0.609 for TimesNet and from 0.306 to 0.642 for KAN-AD. A validation-selected KAN-AD configuration reaches mean F1 0.795 across three seeds on the existing test population. On fresh post-selection sequences, KAN-AD reaches F1 0.793 versus 0.732 for the fixed mean-change rule; in a higher-noise condition it reaches 0.646 versus 0.367. This experiment demonstrates why a dynamic-event benchmark evaluates both the score model and the rule that turns scores into intervals. It also supplies event-level precision, recall and F1 alongside matched-structure agreement. These synthetic localization comparisons evaluate how a

score becomes an event interval (Fig. 11). The separate real-window human and AI reference panels, their source-specific agreement and detector comparisons are provided in Supplementary S6 and S25.

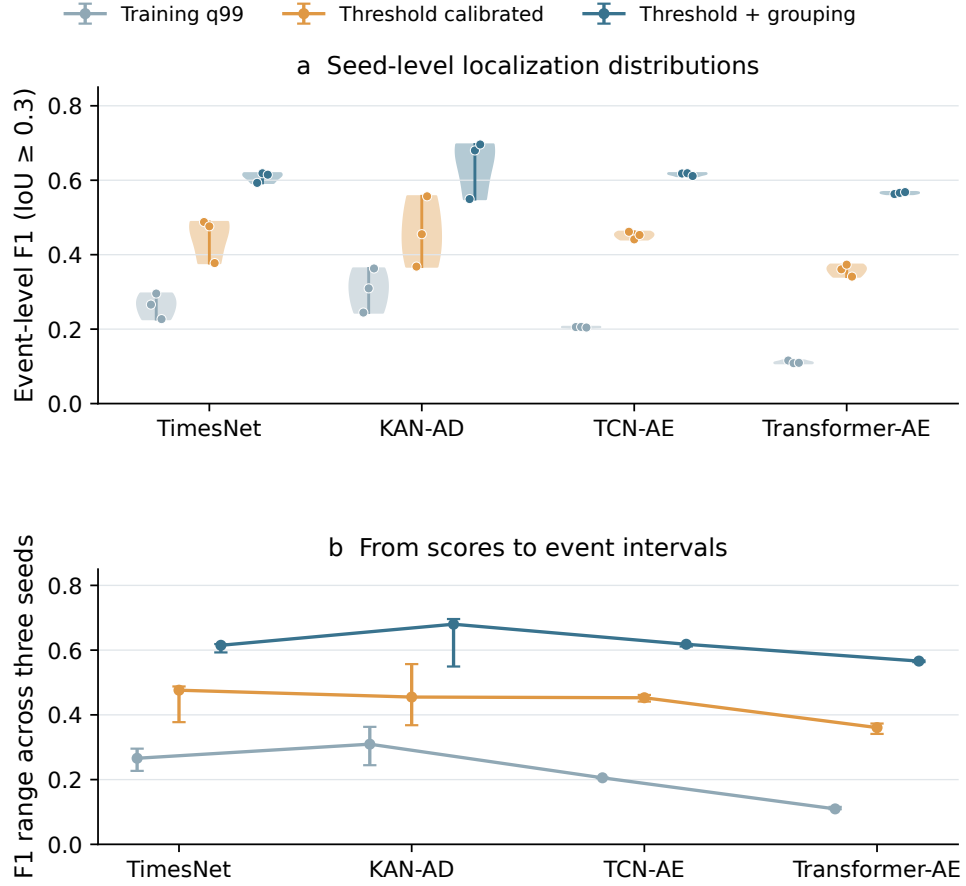


Figure 11: Event-level localization after score calibration and interval grouping. The upper panel shows the three archived seed values as a compact violin and point distribution; the lower panel compares the corresponding median and min-max range across seeds. The grouping rule is evaluated at a fixed IoU threshold, so it measures the complete score-to-interval interface rather than the anomaly score alone.

2.8. *Linking historical shape to rule-based forecast costs*

The operational test translates forecast quality into an explicit monetary outcome. We reconstructed 15-min target points from the 33-turbine Pizhou minute archive and applied the ultra-short-term accuracy component of Jiangsu’s 2022 No. 53 rule [29]. For the provider-reported 87.45-MW farm, each forecast outside the admissible error band incurs CNY 34.98: the band is 3% of capacity at 15 min and 13% at 4 h. This rule-based simulation evaluates every eligible test point, so both avoided and newly introduced charges enter the total.

The information design connects the representation analysis to an operational clock. Completed two- or four-hour power prefixes are encoded as 25-point trajectories, raw principal components or angular components with protected polarity. These causal prefixes contain only pre-issue observations; they are distinct from the event-centred shapes used for physical interpretation. The longer-horizon experiment adds archived Japan Meteorological Agency (JMA) and Global Forecast System (GFS) fields with a fixed 24-h forecast lead [30]. Validation selects representation, weather inputs and the output action of a common probability-forest model. The selected 15-min pipeline uses raw/PCA6 from completed power history; the four-hour pipeline combines protected-polarity angular coordinates with JMA/GFS forecasts.

The selected pipelines reduce both error and accuracy charges on the chronological test period (Fig. 12). At 15 min, charges fall from CNY 112,705.56 to CNY 108,472.98, a saving of CNY 4,232.58 or 3.76% (95% seven-day block interval 2.41–4.96%). At four hours, charges fall from CNY 148,804.92 to CNY 121,695.42, saving CNY 27,109.50 or 18.22% (15.43–21.38%). Four-hour nMAE falls from 17.557% to 14.167%. The corresponding test populations contain 9,450 and 9,345 forecast targets over 16 occupied seven-day blocks. With transformations and selected settings fixed, three classifier seeds retain four-hour fee reductions of 17.51–18.22%. Thus the gain persists across both calendar resampling and classifier randomness.

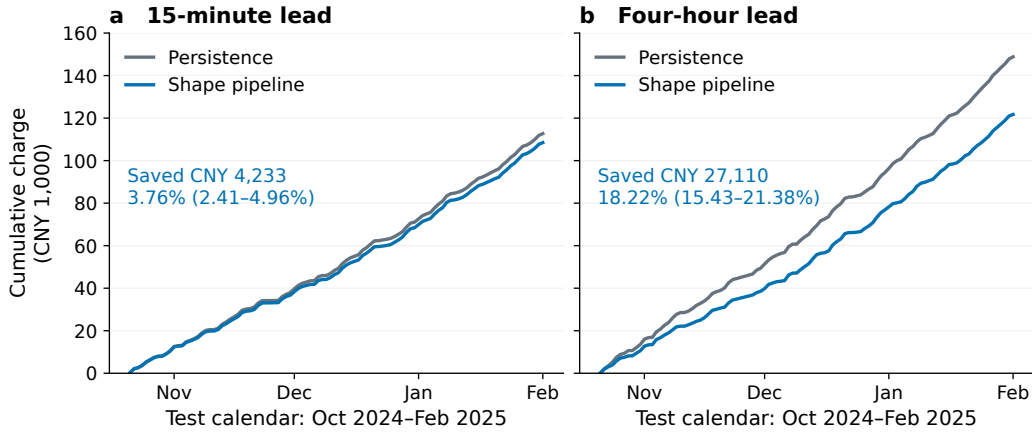


Figure 12: Cumulative simulated accuracy charges on the chronological Pizhou test calendar. Panels a and b compare persistence with the validation-selected shape pipeline at 15 min and four hours, respectively; all eligible points enter each curve. Charges use the 2022 Jiangsu ultra-short-term accuracy component at the provider-reported 87.45-MW capacity. Inset text reports full-period savings and paired seven-day block 95% intervals. The test contains 9,450 targets at 15 min and 9,345 at four hours.

The comparison table places the shape pipeline alongside scalar/weather and nMAE-selected alternatives under the same test targets and capacity. All three learned approaches deliver lower charges than persistence. The 18.22% primary result describes the complete forecast pipeline, encompassing historical information, issued weather and forecast selection. Supplementary Tables S59–S62 retain the information sets, matched control differences and full uncertainty analysis.

Lead	Validation-selected approach	nMAE (%)	Accuracy charge (CNY)	Saving (CNY)
15 min	Persistence	3.039	112,705.56	0.00
15 min	Shape pipeline	2.919	108,472.98	4,232.58
15 min	Scalar + weather	2.932	108,787.80	3,917.76
15 min	nMAE-selected reference	2.910	106,654.02	6,051.54
4 h	Persistence	17.557	148,804.92	0.00
4 h	Shape pipeline	14.167	121,695.42	27,109.50
4 h	Scalar + weather	14.179	121,555.50	27,249.42
4 h	nMAE-selected reference	13.589	121,100.76	27,704.16

Large changes explain where the four-hour economic opportunity is concen-

trated. The 3,079 targets whose change from the persistence value reaches 20% of capacity constitute 32.95% of the test population but contribute 72.38% of baseline charges. The selected pipeline saves CNY 19,134.06 within this high-ramp population. Its full-calendar saving also accounts for improved moderate changes and charges introduced at previously admissible points (Supplementary Table S63). This links event-scale variability to a monetary outcome while retaining the full operating population.

A fixed-model action experiment isolates why average error and cost can rank forecasts differently (Fig. 13). With the same four-hour raw25 probability forest and persistence blend, choosing the centre of the most probable admissible band instead of the conditional mean reduces charges from CNY 139,675.14 to CNY 124,948.56. The saving is CNY 14,726.58, or 10.54% (seven-day block interval 4.55–16.05%), even though nMAE changes from 15.728% to 16.769%. The probability model, input information and test targets are held fixed. The change is the action extracted from the predictive distribution: the conditional mean targets squared error, whereas the band-centred action targets the number of charged exceedances.

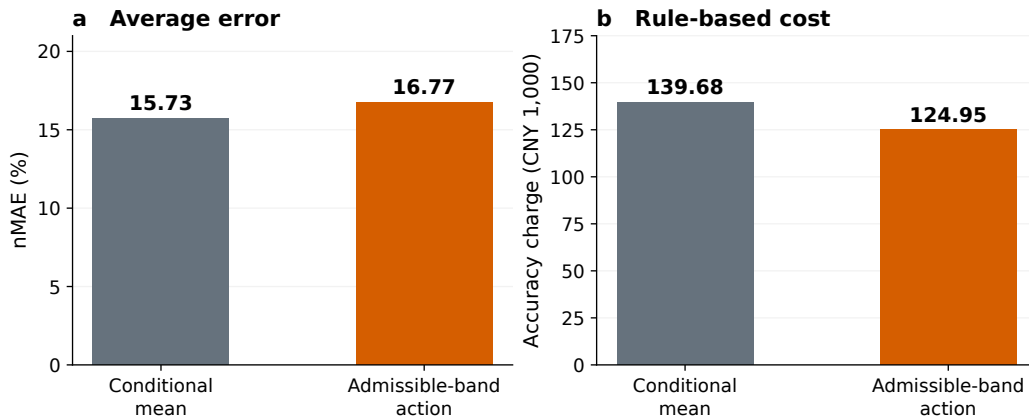


Figure 13: A fixed probability model supports different forecast actions. At the four-hour lead, the raw25 model without numerical weather forecasts and its 0.75 persistence blend are held fixed. The conditional-mean action yields lower nMAE, while the admissible-band action yields lower simulated accuracy charges. The charge reduction is 10.54% with a seven-day block 95% interval of 4.55–16.05%. This is a same-model action ablation, separate from the globally selected pipelines in Fig. 12 and the comparison table.

A complementary British storage simulation tests operating value after an explicit cycling charge. Over 91 complete days in the first half of 2021, the forecast-only event-information controller with battery power equal to 10% of farm capacity

and two-hour energy capacity gains GBP 5,986.77 relative to no storage, after a GBP 10/MWh throughput charge. The market-information comparator gains GBP 7,168.89 under the same asset constraints. These are net operating gains before capital costs; the complete capacity, control and trading comparisons are retained in Supplementary S23.

The decision result is therefore twofold: available shape and weather information support a fee-saving forecast pipeline, and the charging rule determines how a forecast distribution should be converted into an action. The supplementary British price and storage experiments provide the complementary settlement and asset-control tests (S17, S19–S20 and S23). Together, these tracks make decision value an explicit evaluation stage rather than an inference from a clustering score or a single average-error metric.

3. Discussion

This study establishes event measurement as a link between power trajectories, physical-process information and operational evaluation. Its central result is constructive: representations can be chosen according to the distinctions that a scientific or operational task requires. The four-layer framework supplies a controlled comparison interface; polarity protection isolates a recoverable encoding ambiguity; and complete wind-process reconstruction reveals a stronger role for temporal geometry in compound events. The resulting contribution is a design principle supported at several levels, from mathematical information preservation to measured atmospheric dynamics and rule-based decisions.

The ordered and angular representations reveal why a single clustering score is an incomplete design objective. Ordered partitions retain polarity strongly. Angular partitions retain substantial conditioned geometry. A protected polarity coordinate combines a useful invariance with an explicitly retained physical distinction. Complete-process reconstruction extends that argument from sign to temporal evolution: the strongest Hill gains occur in power-defined compound events and remain after duration standardization. Across both instrumented farms, the improvement in internal wind geometry identifies useful information beyond an endpoint trend. This suggests a complexity-aware interface: compact summaries describe coarse changes, while ordered, scale-aware trajectories preserve the reversals needed to resolve compound dynamics. The nearest-prototype margin supplies a complementary local criterion for assignment stability.

Coarse classes and calibrated probabilities serve different roles. A cluster label is a useful summary of a recurring trajectory, while a physical-process prob-

ability quantifies what that trajectory suggests about an external measurement. In the five-site tests, adding a hard four-class label to scalar metadata gives small, heterogeneous incremental gains. Continuous event information is richer. The source probabilities also require field calibration when transferring from regional reanalysis to local LiDAR. Together these observations motivate an event interface containing shape, metadata, prototype membership and calibrated task probabilities, with each component evaluated for its intended use.

The economic analysis completes this task-aware argument. A historical trajectory becomes operationally useful through an issued forecast and an applicable loss function. Under a documented accuracy-charge rule, the selected four-hour pipeline reduces both average error and monetary charges, with gains concentrated in a calendar where large changes account for most baseline charges. The fixed-model action experiment isolates a second principle: preserving a predictive distribution allows the charging rule to select the forecast action. The useful output can therefore differ from the output that minimizes average error. Structural agreement, physical discrimination and monetary performance are complementary tests of a representation’s purpose.

3.1. Limitations and scope conditions

Structural conclusions refer to matched, encodable events, with their coverage and support reported. Regional-weather and LiDAR relationships establish external association and probabilistic discrimination; causal effects of weather or operation require a supported intervention design. Hill control records and SCADA-based yaw indicators offer routes to mechanism analysis, with control execution, transition periods and wake interference requiring separate checks. Greece and SDWPF contribute structural tests; independent weather matching awaits verified absolute clock and location information.

The auxiliary real-window reference contains three human observers and seven AI review sessions, with source-specific agreement reported in Supplementary S25. AI consistency reflects a shared model-based assessment process; event-boundary errors and onset delay require corresponding labels. The 2026 LiDAR calibration and subsequent trajectory-reconstruction task are exploratory chronological tests on previously studied periods. The reconstruction task has four occupied seven-day blocks in each test population; T11 and T07 share one farm’s weather, and SMARTEOLE adds a second farm under its provider’s operating-period filter. Complete event and post-event context support retrospective physical reconstruction. Operational use requires an issue-time predictor, as evaluated separately in the forecast-to-cost track. Duration standardization and coordinate

perturbations examine mechanisms within these data; additional periods and instrumented sites will test how the complexity-dependent benefit transfers.

The Jiangsu calculation is a replay of the 2022-version accuracy component, with capacity taken from provider metadata. Actual invoices also require applicable amendments, submitted forecasts, exemption and reporting records. The newly added forecast task uses a previously studied archive, and the weather inputs use an administrative-area proxy rather than verified turbine locations. Scalar/weather and nMAE-selected controls achieve comparable or lower charges in the present test, so the demonstrated monetary gain belongs to the full pipeline; the independent contribution of morphology requires further confirmation. The British experiments separately evaluate gross exposure, signed settlement and operating revenue under their recorded prices and constraints (Supplementary S23). Their outcomes depend on the contract, decision and storage objective. France’s RTE rules define a transferable settlement calculation [31]; evaluation on La Haute Borne requires aligned historical prices and a specified balance-responsible-party perimeter.

The framework gives the community a concrete reporting and benchmark target. An event result should carry its physical-time protocol, matched population, task-relevant information and decision horizon. Benchmark tracks should separately score structural survival, event localization, independent-process tracking and forecast-to-cost performance. This separation rewards measurements that are useful for a stated scientific or operational purpose, and provides a reusable route from event discovery to physical interpretation and dispatch research.

4. Methods

4.1. Data and common observation conditions

The study includes Pizhou (33 turbines), Suining (14), Yandun (117), La Haute Borne (4), Hill of Towie (21), Greece (10) and SDWPF (134). Chinese archives contribute 298 turbines and European archives 35. La Haute Borne is associated with the OpenOA data ecosystem [32]; the Greek and SDWPF archives supply independent public monitoring resources [33, 34]. The main source periods are September 2023–February 2025 for Pizhou, September 2023–August 2024 for Suining, February–August 2024 for Yandun, 2014–2015 for La Haute Borne, 2020 for Hill of Towie, January–June 2020 for Greece and 2020–2021 for SDWPF. Hill of Towie 2021 January–June supplies the frozen temporal evaluation, and its 2026 measurements supply the instrumented physical test [35]. ERA5 denotes the European Centre for Medium-Range Weather Forecasts fifth-generation reanalysis.

Complete native observations form each half-hour average. Explicit missing bins separate valid runs. SCADA interval-end timestamps at Hill of Towie are shifted to interval starts. A chronological 60/20/20 split defines fitting, validation and test periods. Main detector normalization uses each turbine’s training-period q99.5 power, expressed as a production scale. The same empirical method is used for Pizhou and Greece; fitted transformations remain separate from evaluated data [36]. Nameplate capacity supplies physical MW units where verified, including the 21×2.3 -MW Hill fleet. Event contexts remain within one split and valid run.

4.2. Protocol space and event hierarchy

Let a protocol be $\pi = (Q, D, M, F, C)$, where Q specifies clock, sampling, quality and amplitude calibration; D constructs intervals; M establishes correspondence; F encodes trajectories; and C assigns structure. A controlled comparison changes an identified component while retaining the remaining conditions. Structural agreement is estimated on the matched, encodable population. Weather-risk and economic estimands use their eligible calendar populations.

The complete experiment map is provided in Supplementary Table S0. It records the data period, analysis unit, purpose and status of each catalogue, transfer, physical-process, forecast and cost experiment, so that the structural, physical and decision estimands remain distinct.

The seventeen detector configurations include fixed-amplitude changes, historical financial tails, adjacent-mean changes, fixed and adaptive corridors, the Swinging Door Algorithm (SDA) and the 2015 conference Optimized Swinging Door Algorithm (OpSDA) formulation [6, 7]. Nominal lags are 60, 120 and 240 min. Historical tails use a 24-h preceding window with a 0.10 minimum amplitude and 0.05/0.95 quantiles. Threshold and mean-shift configurations use 0.20 relative power in the main analysis. The adaptive corridor sets and freezes its width at each segment start from available past differences. Composite intervals join adjacent opposite legs, retain child identifiers and record a measured internal turning point, with a four-hour primary extent.

4.3. Correspondence and structure scores

For positive-duration intervals $I_a = [s_a, e_a)$ and $I_b = [s_b, e_b)$, let $J = \max(0, \min(e_a, e_b) - \max(s_a, s_b))$ and $U = (e_a - s_a) + (e_b - s_b) - J$. The intersection over union (IoU) is

$$\text{IoU}(I_a, I_b) = J/U, \quad U > 0.$$

The implementation rejects zero-duration and reversed intervals ($e \leq s$) before overlap calculation. A zero union is an invalid comparison and receives no score or edge. Empty catalogues contain no matches and their coverage is undefined; intervals touching only at an endpoint have $\text{IoU}=0$. For overlapping valid intervals, U equals the enclosing duration used by the implemented candidate-edge calculation. Matches share site, turbine, split and hierarchy. Greedy assignment sorts feasible edges by descending IoU with deterministic tie rules, using $\text{IoU} \geq 0.5$. If N_a and N_b are the eligible event counts and M is the number of one-to-one matches, left and right coverage are M/N_a and M/N_b . Primary summaries use supported, informative configuration pairs common to all six representations and three seeds, first taking the median across seeds within each pair and then across pairs. ARI [37] and normalized mutual information (NMI) quantify partition agreement. Constant partitions and pairs with fewer than 100 matches remain in separate support tables.

The many-to-many extension builds a bipartite graph for each site, turbine, test split and configuration pair. All edges satisfying $\text{IoU} \geq 0.5$ are retained, with 0.3/0.7 sensitivities. Connected components classify one-to-one, one-to-many, many-to-one and many-to-many correspondence. A primitive-only arm keeps the original event population; a hierarchy-inclusive arm allows primitive intervals to correspond to composite V/inverted-V episodes. Coverage counts unique connected nodes on each side, never edge multiplicity. Each component receives one vote when comparing label composition: if p_a and p_b are its four-class frequency vectors, composition overlap is $\sum_j \min(p_{aj}, p_{bj})$. This descriptive score preserves mixture proportions, while within-component temporal order is represented by the stored intervals and turning points. It is distinct from ARI. A separate containment sensitivity uses intersection/minimum duration ≥ 0.5 , allowing a short leg to belong to a longer episode. Components longer than four hours remain explicitly counted as linked measurement supports. Shared seven-day calendar blocks anchor each component at its earliest start; 2,000 draws quantify conditional temporal variation. Per-event predictions use the frozen Pizhou raw25 k=4 seed-41 model.

Configuration names fix the left/right order lexicographically. A supported component pair contains at least 100 connected components; this count is descriptive and does not filter the all-configuration site summary. Site composition scores and topology proportions weight components equally across all 136 configuration pairs. Pooled coverage divides total unique connected nodes by total eligible nodes across those comparisons; an event can contribute once within each comparison. The main one-to-one coverage table instead averages the 136

configuration-specific fractions equally. An exact count reconciliation confirms identical primitive populations and one-to-one matches under both summaries (coverage_aggregation_reconciliation.csv).

4.4. Representations, clustering and information preservation

A shape has four pre-event coordinates, seventeen interpolated internal coordinates and four post-event coordinates. Let p_j denote these sampled power values and p_s the event-start power. We formed $x_j = (p_j - p_s)/a$ with $a = \max_j |p_j - p_s|$ over all 25 coordinates, requiring $a > 0$ and finite values. Hence x belongs to $[-1, 1]^{25}$; zero-variation supports were ineligible for shape analysis. This per-event normalization retains sign and temporal geometry, while amplitude, duration and initial power remain explicit metadata. Context locations preserve physical time across sampling grids. The six representations are raw25, statistics9, raw/PCA6, GAF/PCA6, signed-channel GAF/PCA6 and GAF/PCA5 plus polarity. K-means at $k=4$ is the primary analysis; $k=2/6$ and Gaussian-mixture/medoid comparisons use the same training budget and frozen test events.

For the finite signed unit-domain vector x , set $c_i = \sqrt{1 - x_i^2} \geq 0$. The Gramian angular summation field (GASF), the GAF variant used here, is $G = xx^T - cc^T$ [20], equivalent to $G_{ij} = \cos(\arccos x_i + \arccos x_j)$. For an exact, uncompressed full field, its diagonal yields $|x_i| = \sqrt{(G_{ii} + 1)/2}$, and $G + cc^T = xx^T$. Choose an anchor r with $|x_r| > 0$; the sign bit $b = \text{sgn}(x_r)$ gives $x_r = b\sqrt{(G_{rr} + 1)/2}$ and $x_i = (G + cc^T)_{ir}/x_r$. The deterministic largest-magnitude anchor, with the first index breaking ties, is recoverable from the diagonal. The all-zero path is already determined, although constant supports are excluded by preprocessing. This recovers the normalized trajectory; restoring absolute power also requires its offset and scale. PCA retains only part of the field, so GAF/PCA5 plus polarity is an empirical six-dimensional representation, not an exact-inversion guarantee. It preserves the anchor sign explicitly and is evaluated on held-out clustering and physical targets.

For fixed Euclidean prototypes, let m be the distance to the second closest prototype minus the distance to the closest. A representation perturbation of norm less than $m/2$ preserves the closest assignment by the triangle inequality. This supplies a local stability condition within a shared representation space.

Conditional information analysis stratifies configuration pair and training-defined physical descriptors. Individual analyses condition on direction, amplitude, duration or pre-window power; the joint analysis combines coarse scale strata and both event directions. Strata require at least thirty pairs. One

hundred within-stratum, within-turbine permutations quantify the chance contribution under the preserved controls. Conditional mutual information, remaining label entropy and retained support are reported together.

For matched-event partition labels L_a, L_b and conditioning stratum W , we calculated

$$I(L_a; L_b | W) = \sum_w p(w) \sum_{a,b} p(a, b | w) \ln \frac{p(a, b | w)}{p(a | w)p(b | w)}.$$

The reported excess is the observed value minus the mean within-stratum permutation value. Descriptor cut points were fitted on Pizhou training events: individual analyses used terciles, and the joint analysis used median splits for amplitude, duration and starting power together with both event directions. All information and permutation calculations used retained matched pairs.

4.5. Physical reference and probability experiments

Hourly ERA5 wind, temperature, pressure and related surface fields provide regional context for five geographically aligned archives [38, 39]. The event-level physical target is $\Delta u = u_{100}(e) - u_{100}(s)$, classified as decrease for $\Delta u \leq -1.5$ m s⁻¹, increase for $\Delta u \geq 1.5$ m s⁻¹, and small net change otherwise; event duration satisfies $0 < e - s \leq 4$ h. The ± 1.5 m s⁻¹ cutoffs are study-defined symmetric contrast levels, fixed for the regional probe before the later LiDAR transfer. They provide a common operational definition of a substantial endpoint wind-speed tendency, rather than a turbine cut-in/cut-out boundary or a universal meteorological ramp threshold. The four-hour limit matches the longest principal detector horizon and the composite-event window used in this study, focusing the probe on intraday dynamics. It does not require every wind change to span four hours: the same endpoint contrast may occur over a shorter event. A V-shaped excursion may return to small net change while retaining substantial internal variation, which is evaluated through event morphology and the overlap analysis. NOAA Integrated Surface Database (ISD) observations are archived with variable-level quality codes and precipitation accumulation duration [40]; independent LiDAR supplies the local-wind test.

Physical probes share training-event identifiers and two gradient-boosted-tree leaf budgets. Training fits the model; validation selects leaf budget and probability temperature. Scalar controls are direction, amplitude, duration, starting power and pre-window mean. Standalone representations, scalar-plus-representation inputs and scalar-plus-hard-cluster inputs form distinct comparisons. Proper log and

Brier scores evaluate probabilities [41]. Log-score differences estimate model-usable predictive information.

The 2026 instrument test uses 58-m horizontal wind from ZX300 unit 2428 near T11 and 208-m fit-derived hub-height wind from ZXTM unit 5060 on T07. Positive packet counts and physical wind values define eligible measurements; diagnostic flags remain archived. Exact 10-min endpoint support and complete intervening wind bins are required. The prescribed -10 , 0 and $+10$ -min wind-clock offsets are all retained. Frozen Hill-2020 heads receive frozen Pizhou representations. A separate multinomial-logistic calibration of source log probabilities uses the chronological field splits specified in Results, with C selected from 0.1, 1 and 10 on validation data.

The SMARTEOLE field archive contains seven turbines with provider-preprocessed 1-min SCADA, a 1-min control log, a ground WindCube profiler and coordinate metadata [42]. The provider removes rows with at least one stopped turbine. We aggregate complete 30-min means without interpolation, freeze the median Pizhou training q99.5 scale, and apply Pizhou transformations and calibrated physical heads without target-site fitting. WindCube 80-m endpoint changes use the same signed three-class target and four-hour duration limit.

The reviewer-triggered factorial extension used the same five scalar descriptors (endpoint direction, signed endpoint change, duration, starting power and pre-window mean), with either no additional input, five angular principal components, polarity alone, or their combination. Every feature arm shared event identifiers, chronological splits and the two-leaf-budget probe search. Angular coordinates came from one frozen Pizhou transform. Five ERA5 sites used their recorded training and validation events; SMARTEOLE received frozen Pizhou heads; LiDAR received Hill-2020 heads followed by the same chronological calibration. Paired block differences were calculated for direction AUROC, three-class log loss and Brier score. Random sign, test-only sign errors, consistent sign reversal, pre-PCA signed channels and alternative anchor coordinates isolate different mechanisms. Supplementary S24 specifies all thirteen feature arms and the noise tests.

4.5.1. Complete wind-process reconstruction

The process experiment retained deduplicated primitive and compound intervals of 60–240 min with at least seven consecutive valid 10-min wind observations. A compound flag was assigned when a retained power-catalogue definition contained opposite legs. Detection boundaries remained fixed on the 30-min grid; paired 10/30-min power tracks shared event identifiers, targets and splits. For event start s , end e and relative phase $r_j = j/16$, the measured target was

$Y_j = u(s + r_j(e - s)) - u(s)$, $j=0, \dots, 16$. Linear interpolation used only continuous valid wind support. Full-trajectory RMSE scored $j=1, \dots, 16$. Internal geometry used $R_j = Y_j - r_j Y_{16}$, with the same operation applied to the predicted trajectory before scoring. This second metric isolates intermediate evolution from endpoint tendency.

Scalar controls described endpoint direction/change, duration, initial and contextual power, amplitude scale, variation, curvature, extreme timing and eight cube-root-power summaries. Each temporal arm received the common expanded scalar inputs. Training-only transforms supplied ordered, principal-component and angular encodings, including scale-restored event trajectories. Each arm compared four ridge penalties and four extremely randomized tree configurations. A scalar-only residual model received the same cross-fitting and correction budget as temporal residual models. Validation MSE selected both the scalar comparator and trajectory pipeline before the test was evaluated; all three selected pipelines used native 10-min scale-restored trajectories and ridge regression. Supplementary S26 and Tables S75–S80 specify the full matrix, split dates, selected controls, event hierarchy and negative controls.

We calculated paired relative RMSE reductions on the same events using 2,000 shared farm-calendar block draws. Compound and primitive-only strata were defined from power records. A second analysis weighted their common duration strata by the smaller group count, then compared their gain difference. This descriptive standardization retained all common-support durations without consulting wind outcomes. Frozen-decoder perturbations preserved scalar inputs and endpoints while either shuffling interior coordinates or reassigning chord residuals between similarly scaled events.

4.6. *Reference panels and forecast-to-cost experiments*

The auxiliary real-window reference retained three human observers and seven independently prompted AI review sessions as separate source types. Each assessed the central two hours of the same four-hour displays. Source-specific agreement, complete prompts, label mappings and the distinction from the controlled localization benchmark are documented in Supplementary S6 and S25.

The primary cost test used Pizhou’s minute records to construct exact quarter-hour target power and inputs available at least one minute before issue. The provider’s capacity note specified 87.45 MW. The archive supplied 49,920 grid points, including 48,206 complete target points. Chronological training and validation ended at 00:00 China Standard Time on 9 July and 21 October 2024; the test continued to 1 February 2025. Training targets and evaluation issue times

respected these boundaries. Normalized mean absolute error (nMAE) was the mean absolute power error divided by capacity.

Two- and four-hour completed prefixes were resampled to 25 coordinates and transformed using training-only raw/PCA6, GAF/PCA6 or GAF/PCA5 plus polarity encoders. Scalar controls described recent power, wind, variation and calendar state. Archived JMA and GFS fields used the Previous Runs API’s fixed 24-h-ahead forecasts, rather than target-time observations [30]. Including hourly interpolation, the nominal forecast vintages predated issue by more than nineteen hours at both evaluated leads. The three weather arms used common validation and test targets; model-specific archive coverage determined their training support. Supplementary S22 specifies fields, source coverage and the distinctions between this causal-prefix track and event-centred encoding.

An ExtraTrees classifier estimated power-bin probabilities from training data using 150 trees, maximum depth 16 and minimum leaf sizes 8 or 32. Bins had width 1% of capacity. We compared the conditional mean, conditional median and the forecast maximizing probability inside the permitted error band, each blended with persistence at weights 0, 0.25, 0.5, 0.75 or 1. Validation selected the minimum failed-point count, with nMAE and then the smaller blend resolving ties. A separately selected nMAE reference used the same candidate pool. Shape transformations and model settings were fixed before testing; classifier seeds 41–43 assessed residual fitting variability.

Jiangsu’s 2022 No. 53 Article 44(II) specified point accuracy $A_t = 100(1 - |P_t - \hat{P}_t|/C)$, with thresholds 97% at 15 min and 87% at four hours [29]. For capacity C in MW and $\tau_h = 0.03$ or 0.13, we calculated the accuracy charge in Chinese yuan (CNY) as

$$F_h = 0.4C \sum_{t \in \mathcal{T}_h} \mathbf{1}\{|P_t - \hat{P}_t| > \tau_h C\}.$$

$\mathcal{T}_h$ contains all eligible test targets at horizon h . Equality to the tolerance qualifies; numerical comparisons used a 10^{-12} normalized tolerance. Each exceedance therefore cost CNY 34.98. This ultra-short-term component has no 2% monthly allowance; that allowance belongs to the separate medium/short-term component. We counted observed eligible points without extrapolating missing targets. Relative savings were $100(F_{\text{persistence}} - F_{\text{model}})/F_{\text{persistence}}$. Paired 3/7/14-day calendar blocks and 2,000 resamples quantified uncertainty. The same-model action test retained the fitted forest, input matrix, blend and target times while changing only the forecast action.

The British observed-price and storage protocols, including the definition of each scheduled position S_t , signed costs, credits, inventory constraints and complete-revenue controls, are preserved in Supplementary S17, S19–S20 and S23 [43, 44]. These evaluate separate decision quantities from the Jiangsu point-accuracy charge.

4.7. *Event-level localization benchmark*

The controlled detection benchmark uses 96-point synthetic trajectories with known event intervals, 1,600 normal training sequences, 400 validation sequences, 800 test sequences and 800 higher-noise test sequences. TimesNet, KAN-AD, a temporal convolutional network (TCN) autoencoder and a positional Transformer autoencoder receive matched inputs and training budgets. Thresholds are selected from training or validation scores, and event grouping varies hysteresis, merge gap and minimum duration. IoU 0.3 is primary, with 0.1 and 0.5 sensitivity checks. Fresh post-selection sequences use fixed generator and checkpoint choices.

4.8. *Uncertainty, software and verification*

The original structural protocol fixed K-means at $k=4$, with $k=2/6$ as sensitivity analyses. Zero-minute LiDAR clock alignment was the primary frozen-transfer alignment, and the Jiangsu forecast pipelines were selected on validation before their new weather-arm test scores. Local LiDAR recalibration, the scalar-controlled factorial, complete-process reconstruction and the AI-reference expansion are exploratory extensions developed after earlier results were available. The process experiment selected models on validation full-trajectory MSE, then evaluated the fixed test population and its power-defined hierarchy; internal geometry and derived process classes are complementary endpoints. Their complete comparisons are reported without significance-based model or reviewer selection; the many exploratory intervals are not simultaneous family-wise intervals.

Shared within-farm calendar blocks provide 2,000 bootstrap draws, with seven days primary and three/fourteen days as sensitivities [45]. Turbines observed in the same sampled block receive the same weight. Comparisons report event counts, supported configuration pairs and occupied blocks. Block intervals describe time-dependent sampling uncertainty; across-configuration interquartile ranges describe protocol variation. Analysis software exposes detection, matching, encoding, clustering and evaluation interfaces, with unit checks for train-only fitting, time windows, correspondence, angular inversion, cashflow units and storage energy conservation. AI-assisted programming support was used under

direct author supervision; the authors independently checked the code, numerical outputs, citations and claims.

5. Data and code availability

Public source archives include Greece (doi:10.5281/zenodo.14546480), SDWPF (doi:10.6084/m9.figshare.24798654), La Haute Borne and Hill of Towie, including the instrumented release (doi:10.5281/zenodo.22662930). Processed-data access, v18–v26 aggregate results, editable figures, manuscript sources and reproducibility scripts are available in the [v0.7.0 study release](#). The economic replication bundle in the preceding v0.6.0 release supplies de-identified transformed inputs, fitted models and six exact-output reproduction cases for the native policy-free task. The current release adds complete-process reconstruction results, scalar-controlled physical contrasts, AI review inputs and responses, and source-separated coincidence counts for reproducing reference agreement. Provider attribution, source terms and the SMARTEOLE licence route accompany the release. Supplementary material specifies the relation between public relative clocks and internal meteorological alignment.

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CRedit authorship contribution statement

Conceptualization, P.G. and C.Y.; methodology, P.G.; software, P.G.; validation, P.G.; formal analysis, P.G.; investigation, P.G.; resources, C.Y.; data curation, P.G.; writing—original draft preparation, P.G.; writing—review and editing, C.Y.; visualization, P.G.; supervision, C.Y.; project administration, C.Y.; funding acquisition, C.Y.

Declaration of competing interest

None declared.

6. Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used ChatGPT (OpenAI) for language editing, manuscript organization, code review and research-record management under author supervision. The authors reviewed and edited the output, independently checked the data, analyses, references and claims, and take full responsibility for the content of the publication.

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